Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/16/2004
PI Name:	Javanainen, Juha D.Sc.		
Project Title:	Quantum Optics Style Theory for Degenerate Bose and Fermi Gases		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06269-3046	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	04/01/2003	**End Date:**	11/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2880		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We are currently pursuing two main lines of research. First, we are developing a quantum optics style theory for dynamical phenomena in dilute quantum degenerate Fermi gases in the neighborhood of a Feshabch resonance. Second, we are studying the dynamics of ultralow-temperature gases in optical lattices. Our goal is to generate new theoretical concepts in the physics of degenerate atomic gases, such as a Bose-Einstein condensate or a degenerate Fermi gas realized in alkali metal vapors. In the usual manner of basic theoretical research, when the concepts and ideas arising from this work circulate in the community, new theoretical ideas, experiments, and ultimately new technologies will follow. Prospective applications of interest to NASA include atomic clocks and deep-space navigation.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	The project has emerged from the start-up phase. We have initiated two new lines of research, and obtained important results in both of them. (i) We have developed a numerical theory for the conversion of a degenerate Fermi gas of atoms into a gas of molecules during a sweep of the magnetic field across a Feshbach resonance. The key conceptual finding is that the conversion is driven by a dynamical instability of the Fermi sea of atoms against molecule formation. Comparisons with existing experiments are satisfactory, and we have made the prediction that the initial temperature of the atomic gas sets the efficiency limit for the atom-molecule conversion. (ii) We are studying the dynamics of a Bose-Einstein condensate of atoms in an optical lattice in the presence of a driving force that pushes the condensate from one lattice site to the next. In semi-quantitative agreement with the experiments, we have found that, paradoxically, the motion of the atoms between the sites becomes chaotic in the limit when the force is small. We have also continued work that started during the predecessor NASA grant NAG8-1428: (iii) When atoms obeying the Fermi-Dirac statistics are put in an optical lattice, a fractional fermion may result; the system behaves as if it had half of an atom in it. We have shown that such a half-fermion may be detected optically. Finally, during the report period several publications have appeared that originate from incidental and/or discontinued lines of earlier research.
Bibliography Type:	Description: (Last Updated: 01/08/2020)
Articles in Peer-reviewed Journals	Javanainen J, Ruostekoski J. "Optical detection of fractional particle number in an atomic Fermi-Dirac gas." Phys Rev Lett. 2003 Oct 10;91(15):150404. PubMed PMID: 14611454 , Oct-2003
Articles in Peer-reviewed Journals	Varro S, Javanainen J. "Gauge-invariant relativistic wigner functions in external electromagnetic fields." Acta Physica Hungarica Series B, Quantum Electronics. 2004 May;20:15–23. https://doi.org/10.1556/APH.20.2004.1-2.5 , May-2004
Articles in Peer-reviewed Journals	Ishkhanyan A, Mackie M, Carmichael A, Gould PL, Javanainen J. "Landau-Zener problem for trilinear systems." Physical Review A. 2004 Apr;69(4):043612. http://dx.doi.org/10.1103/PhysRevA.69.043612 , Apr-2004
Articles in Peer-reviewed Journals	Mackie M, Dannenberg O, Piilo J, Suominen K-A, Javanainen J. "Raman photoassociation of Bose-Fermi mixtures and the subsequent prospects for atom-molecule Cooper pairing." Physical Review A. 2004 May;69(5):053614. http://dx.doi.org/10.1103/PhysRevA.69.053614 , May-2004
Articles in Peer-reviewed Journals	Javanainen J, Kostrun M, Zheng Y, Carmichael A, Shrestha U, Meinel PJ, Mackie M, Dannenberg O, Suominen KA. "Collective molecule formation in a degenerate Fermi gas via a Feshbach resonance." Phys Rev Lett. 2004 May 21;92(20):200402. PMID: 15169331 , May-2004
Articles in Peer-reviewed Journals	Shevchenko A, Kaivola M, Javanainen J. "Thermodynamics of a multicomponent atom sample in a tightly compressed atom trap." Physical Review A. 2004 Jul;70(1):011403. http://dx.doi.org/10.1103/PhysRevA.70.011403 , Jul-2004
Articles in Peer-reviewed Journals	Mackie M, Härkönen K, Collin A, Suominen K-A, Javanainen J. "Improved efficiency of stimulated Raman adiabatic passage in photoassociation of a Bose-Einstein condensate." Physical Review A. 2004 Jul;70(1):013614. http://dx.doi.org/10.1103/PhysRevA.70.013614 , Jul-2004
Presentation	Javanainen, J. "Symbolic calculations in the theory of atoms" Annual Meeting of the American Chemical Society, Philadelphia, August 22-26, 2004 Aug-2004
Presentation	Zheng, Y.; Kostrun, M., and Javanainen, J. "Time-dependent perturbation theory of a Bose-Einstein condensate in a tilted optical lattice" DAMOP 2004, Tucson, AZ, 4/25/04 - 4/29/04 May-2004
Presentation	Javanainen, J., and Mackie, M. "Collisionless conversion of a degenerate Fermi gas into molecules in a Feshbach resonance" DAMOP 2004, Tucson, AZ, 4/25/04 - 4/29/04 May-2004
Presentation	Javanainen, J., and Mackie, M. "Collisionless conversion of a degenerate Fermi gas into molecules in a Feshbach resonance" DAMOP 2004, Tucson, AZ, 4/25/04 - 4/29/04 May-2004
Presentation	Javanainen, J. "Conversion of a degenerate Fermi gas into molecules in a Feshbach resonance" 2004 NASA/JPL Workshop on Physics of Planetary Exploration, Solvang, CA 4/20/04 - 4/22/04 Apr-2004
Presentation	Javanainen, J. "Atom-molecule Bose-Einstein condensate" Colloquium at the Department of Physics and Astronomy, Penn State, 09/11/03. Sep-2003
Presentation	Javanainen, J. "Modeling coherent association of fermionic atoms into molecules" 34th Winter Colloquium on The Physics of Quantum Electronics, Snowbird, UT, 01/04/04 - 01/08/04 Jan-2004